AI Usage

- OpenCode. Big Pickle, OpenCode Zen, accessed during the July 25, 2026 session, prompt: “tell me how the animation works with Player class”, used to create a new Enemy object based on the Player class and apply animations to it.
- OpenCode. Big Pickle, OpenCode Zen, accessed during the July 27, 2026 session, prompt: “can we check which part of an @src/Game/Objects/Enemy/Enemy.cpp is in contact by add another hitbox in front/on top of it or by smth else”
- OpenCode. Big Pickle, OpenCode Zen, accessed during the July 28, 2026 session, prompt: “should we separate check contact in each object like player, enemy or should we combine in one by b2contact and hardcode every case”
- OpenCode. Big Pickle, OpenCode Zen, accessed during the Aug 03, 2026 session, prompt: “make a plan to refactor code and find redundant codes in World/”
- OpenCode. Big Pickle, OpenCode Zen, accessed during the Aug 31, 2026 session, prompt: “can you look into @src/Game/Camera.cpp and @include/Game/Camera.h both in this main branch and in fix/settings-scene, i want to revert both files to fix/settings-scene”
- OpenCode. Big Pickle, OpenCode Zen, accessed during the Aug 31, 2026 session, prompt: “i want to remove the feat camera bounding so delete what relevant to that also”
- OpenCode. Big Pickle, OpenCode Zen, accessed during the Aug 31, 2026 session, prompt: “fix it”
- OpenCode. Muse Spark, opencode/muse-spark-1.2-contributor-free, accessed during the Aug 31, 2026 session, prompt: “if you design a class diagram for this whole project, how would you do it, i think we should separate it into small parts. Also we need to explain the design pattern and the reasoning of our design”
- OpenCode. Muse Spark, opencode/muse-spark-1.2-contributor-free, accessed during the Aug 31, 2026 session, prompt: “Category Feature Points Player Input, Movement and Collision Move 0.25 Jump 0.25 Shoot fire 0.25 Interact with block by jumping 0.25 Break block/pipe in MegaState 0.25 Teleport by pipe 0.25 Enemy Behaviour Goomba 0.25 Piranha Plant 0.25 Koopa 0.25 Koopa revives from their shells after some time 0.25 Pickup/Throw Koopa shell 0.25 Kick Koopa shell 0.25 Items Super Mushroom 0.25 Fire Flower 0.25 1-Up Mushroom 0.25 Super Star 0.25 Mega Mushroom 0.25 Checkpoint flag 0.25 Game Stages Level 1 0.25 Level 2 0.25 Level 3 0.25 Minigame PvP 0.25 Minigame vs AI 0.25 Sounds, Effects and Backgrounds Sounds 0.25 Effect 0.25 Parallax background 0.25 OOD Encapsulation 0.25 Abstraction 0.25 Inheritance 0.25 Polymorphism 0.25 Design Pattern Factory method 0.25 Singleton 0.25 Strategy 0.25 Command pattern 0.25 Decorator 0.25 Additional Requirements AI 0.25 Mutiple players 0.25 Save/load game 0.25 Custom map 0.25 Total 39 9.75

did i miss something for the list of features for my project"

- Muse Spark. muse-spark-1.2-contributor-free, Meta, chat.opencode.ai, prompt: "expand the width of buttons in settings panel to the width of music+keybind+devmode+back", used to widen sub-menu rows to top-row span (890).
- Muse Spark. muse-spark-1.2-contributor-free, Meta, chat.opencode.ai, prompt: "change music and sound volume to bar slider?", used to request visual bar for volumes.
- Muse Spark. muse-spark-1.2-contributor-free, Meta, chat.opencode.ai, prompt: "can we fix the keybind to scroll instead of toggle, and remove the toggle button, separate 2 player keybind by text", used to request scrollable dual-player keybind list.
- Muse Spark. muse-spark-1.2-contributor-free, Meta, chat.opencode.ai, prompt: "shorten the height of the scroll panel, move it down a little bit", used to adjust keybind viewport size and position.
- Muse Spark. muse-spark-1.2-contributor-free, Meta, chat.opencode.ai, prompt: "remove the button for bar slider, just pure bar slider, no rectangle button outside of it. make music volume: x% a text on top of the slider, sound is the same. Music enabled is also a text, the toggle button is just the toggle, no button outside of it. Also do it with dev mode", used to request pure controls without outer rect.
- Muse Spark. muse-spark-1.2-contributor-free, Meta, chat.opencode.ai, prompt: "margin music/sound volume label to the left like others", used to left-align volume labels to 16px.
- Muse Spark. muse-spark-1.2-contributor-free, Meta, chat.opencode.ai, prompt: "can we remove redundant code in bar slide and settings panel", used to request deduplication via UIHelpers.
- Muse Spark. muse-spark-1.2-contributor-free, Meta, chat.opencode.ai, prompt: "can we enable mouse in save game scene without violate anything?", used to ask safety of enabling mouse.
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: "why is <SFML/Graphics.hpp> not found by clangd in AnimationFrame.h but it is ok in other files", used to diagnose clangd include-path and compilation-database configuration.
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: "In included file: 'Enable C++17 or newer for your compiler (e.g. -std=c++17 for GCC/Clang or /std:c++17 for MSVC)'", used to troubleshoot conflicting C and C++ language-standard flags.
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: "why i initialised Player.cpp from a spritesheet correctly but the program is still drawing the fallback rectangle", used to investigate sprite initialization and fallback rendering.
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: "so for it to work properly, i have to scale it by the size of the spritesheet?", used to clarify the separation between spritesheet size, frame size, hitbox size, and rendered size.

- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: “plan how should i refractor the animation system”, used to design separate responsibilities for animation data, playback timing, sprite updates, and rendering.
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: “read the whole animation system, what should i do to refractor it”, used to review the completed animation architecture and identify remaining duplication. Animation Completion and Transitions
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: “read the animator, now i have switched the clip to the default each time a non-looping one has completed, but somehow it is not functioning properly”, used to investigate non-looping animation completion and transitions back to default clips.
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: “now the death animation only play once, and later deaths are instant”, used to diagnose stale animator completion state when replaying non-looping animations.
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: “now i want to preload animation for mario and luigi, how should the design be”, used to design reusable named animation presets.
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: “but if i need to add goomba or kooba, using the same position wont work”, used to ensure animation presets could support characters with different spritesheet layouts.
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: “is each object inheriting animatable on their own instead of gameobject a better move?”, used to compare optional visual inheritance with making every GameObject animatable.
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: “but it will be easier to use the animatable constructor rather than redefining the same thing for all concrete object class”, used to evaluate constructor reuse for shared visual initialization. Input and Player Controllers
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: “read the Game/Behaviours/UserInput and report how to properly use it”, used to evaluate the original input-handling design.
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: “If you want a clean input architecture, use InputManager as a separate controller layer, not directly inside GameObject. how to do this”, used to design a controller layer that converts input events into gameplay commands. Controllable and UserInput Responsibilities
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: “should i put bool _movingLeft = false; bool _movingRight = false; bool _wantJump = false; and void startMoveLeft(); void stopMoveLeft(); void startMoveRight(); void stopMoveRight(); void jump(); in controllable?”, used to decide where movement intentions and action methods should reside.
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: “why

the design ActionStart and ActionEnd is so weird, where im i even supposed to pass the event", used to clarify event routing and remove awkward coupling between SFML events and gameplay actions.

- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: "read the new state of the project, now i want to change behaviours to a vector so that i can add/remove them from a game object, what is a good plan to do this?", used to design runtime composition using a behaviour vector.
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: "why was behaviour.h designed that way", used to understand behaviour lifecycle hooks, ownership, polymorphism, and owner references. Behaviour Ownership and Access
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: "then each concrete behaviour has a behaviour subobject?", used to understand virtual inheritance and base-class storage costs.
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: "why is getBehaviour returning a araw ptr", used to understand non-owning access to behaviours stored with unique_ptr. Box2D Contact Handling
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: "this is how to add contact event listener to the box2d 3.1.0, with the current state of the program, should i retrieve the contact at the GameWorld level and add a system to handle them separately?", used to place contact collection in the physics/world layer while keeping gameplay reactions separate.
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: "how im i supposed to know what kind of contact is it, do i have to work with low level code", used to design contact classification using categories, shape metadata, and object references.
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: "teach me sensor", used to understand Box2D sensors and non-solid overlap detection.
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: "does b2ContactEvents and b2SensorEvents work the same way", used to compare ordinary contact events with sensor begin/end events. Grounding Stability
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: "read my code, check why the feetsensor counter is stuck at 0 for the player", used to diagnose missing feet-sensor events.
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: "ok the main problem is the flickering between grounded and airbone, but it still happens even tho i commented the changing velocity, read the createFeet to see if theres any problem, do not touch any code", used to investigate unstable grounding across tiled floors. Hitbox and Debug Rendering
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: "should the function updateSpriteLayout and drawFallbackRect in the protected of gameobject? why", used to decide the visibility of sprite-layout and debug-rendering helpers.
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: "im not sure the rendering is correct because the objects interact even tho the

hitbox does not indicate so", used to compare Box2D collision geometry with rendered debug hitboxes.

- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: "crash when exit the ingame scene", used to investigate destruction order and invalid resource ownership during scene transitions.
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: "ingame scene crashes when open", used to diagnose invalid animation state or object initialization during scene loading.
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: "how to edit the opacity of the scene", used to explore translucent game-over overlays.
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: "is just drawing it to the ingame scene a better idea?", used to compare a separate game-over scene with an overlay rendered by InGameScene.
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: "how should i add victory and lose animation to the player when the game ends? play the animation over the overlay, or make him celebrate/cry before showing screen?", used to choose a reaction-first end sequence.
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: "plan please", used in context to plan separate reaction and overlay phases for victory and defeat.
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: "read current state before plan to add the win game flag pole", used to inspect the object factory, spawn data, scoring, and scene flow before designing the flagpole.
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: "the flagpole is currently 128x896, can you make its actual hitbox a little smaller but keeping the visual 128x896", used to separate the flagpole's trigger hitbox from its rendered dimensions.
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: "look at my invincible behaviour, why when attached to the player, it does not allow me to phase through enemy", used to distinguish damage immunity from collision-filter changes.
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: "yeah its a temporary invincible, because i want some time of invincible when the player changes form to normal", used to define temporary post-hit invincibility requirements.
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: "the ghostShader does not activate when tempInvicible is on", used to diagnose communication between temporary invincibility and visual shader state.
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: "when invincible is removed from player in update simulation, the player enters airbone, is this a problem from GroundTracker", used to investigate interactions between runtime behaviour removal and grounding state.
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: "plan to add winning condition in minigame mode and change the breakable brick

in the mode to unbreakable, also create a map that consist of only a flat floor in AI mode", used to design last-player-standing results, unbreakable arena terrain, and a flat AI arena. Heuristic VS AI

- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: "currently the heuristic AI is really hesitant when it comes to map corner, how would you suggest fixing that", used to improve arena-edge recovery.
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: "because the characters now have acceleration, in the VS AI minigame mode, the play stands still and the ai just go back and forth, cannot end the game, think of how to solve it", used to identify momentum-induced AI oscillation and attack failures.
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: "now can you add the traction and acceleration back, but change the heuristic so that it uses more information from the players for the calculation (relative acceleration, traction, velocity, position . . .)", used to design kinematics-aware observations and predictive movement.
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: "i suggest using a similar idea to PID controller?", used to explore feedback-control techniques for reducing AI overshoot.
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: "move to Player.h and Player.cpp, edit the code 'float moveSpeed = _baseMoveSpeed; float jumpSpeed = _baseJumpSpeed; if(_character != "mario") { moveSpeed = *0.9f; jumpSpeed* = 1.1f; }', add more base attributes", used to define separate top-speed, acceleration, traction, and jump characteristics for Mario and Luigi.
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: "the base acceleration is too low", used to tune acceleration for more responsive player movement.
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: "can the current settings menu be reused effectively? dont implement anything", used to assess whether the existing settings UI could be embedded during gameplay.
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: "then plan to refractor it to be reusable ingame", used to design a reusable settings panel and fully freezing pause overlay.
- ChatGPT. GPT-5.5 Thinking, OpenAI, chat.openai.com, prompt: "look at `D:\0. APCS\4. Group Projects\CS202_GameProject\assets\guis\Square_premade_buttons_16x16` i want to add a pause button at the right corner of the screen, where the button is the texture starting from 256,0 with size 16x16, it top-left corner becomes 256,16 on hover, clicking it acts the same way as pressing esc", used to specify the pause button's atlas cells, position, hover state, and behavior.
- Codex. 5.6 Luna, OpenAI, prompt: "when the enemy touch the pipe, it will stuck there without changing direction", used to fix enemy collision behavior.
- Codex. 5.6 Luna, OpenAI, prompt: "the distance that enemies detect the

pipe is a little bit long", used to reduce enemy pipe-detection distance.

- Codex. 5.6 Luna, OpenAI, prompt: "I want the enemies to change the direction right after hitting the pipe", used to make enemies reverse immediately after pipe contact.
- Codex. 5.6 Luna, OpenAI, prompt: "add a horizontal pipe into map1", used to add a horizontal pipe to Map 1.
- Codex. 5.6 Luna, OpenAI, prompt: "make the piranha plant get out and get in the piranha pipe", used to animate piranha plants entering and leaving pipes.
- Codex. 5.6 Luna, OpenAI, prompt: "make it dive deeper a little bit so that there is no hitbox out of the pipe, also when it raise up, make it be entirely on top of the pipe", used to correct piranha-plant positioning and hitboxes.
- Codex. 5.6 Luna, OpenAI, prompt: "also, in map 1, erase the piranha plant and replace it with a STATIC Piranha pipe - The Piranha stay there without moving", used to add a static piranha pipe.
- Codex. 5.6 Luna, OpenAI, prompt: "Add warp pipes. When move into a warp pipe, player will teleport to another pipe. Also, add ability for Player to use the warp pipe - both vertical and horizontal pipes", used to implement warp pipes.
- Codex. 5.6 Luna, OpenAI, prompt: "add 2 warp pipes in map 2", used to add two warp pipes to Map 2.
- Codex. 5.6 Luna, OpenAI, prompt: "how to use the warp pipe?", used to explain warp-pipe controls.
- Codex. 5.6 Luna, OpenAI, prompt: "add another warp pipe that has no piranha plant", used to configure pipes without piranha plants.
- Codex. 5.6 Luna, OpenAI, prompt: "what if there are more than 2 warp pipe?", used to support multiple warp pipes.
- Codex. 5.6 Luna, OpenAI, prompt: "add warpTarget and warpID", used to add warp destinations and identifiers.
- Codex. 5.6 Luna, OpenAI, prompt: "update it in the map1, map2, spawns json files too", used to update level data for warp pipes.
- Codex. 5.6 Luna, OpenAI, prompt: "add keybind settings to get down into the pipe", used to add downward pipe controls.
- Codex. 5.6 Luna, OpenAI, prompt: "Currently, the logic of the spawning object and blocks are so trash. Refactor the logic: Split it into 2 layers: 1. Tile Layer… 2. Object Layer… Also, use the prefab pattern too.", used to redesign map and object spawning.
- Codex. 5.6 Luna, OpenAI, prompt: "change the minigames map too", used to apply the map-data refactor to minigames.
- Codex. 5.6 Luna, OpenAI, prompt: "The star is falling through the blocks", used to fix star collision.
- Codex. 5.6 Luna, OpenAI, prompt: "I think we should merge both into one - titleLayer and objectLayer - since when there are too many objects, it is really hard to imagine… ", used to consolidate map layers and pipe metadata.

- Codex. 5.6 Luna, OpenAI, prompt: “change the maps’ data too”, used to migrate existing map files.
- Codex. 5.6 Luna, OpenAI, prompt: “delete “G” with “ground” and replace it with “#” and brick”, used to simplify map symbols.
- Codex. 5.6 Luna, OpenAI, prompt: “review the changed things such as ObjectKind::Tile, the extra collision inside the Block.cpp, ... If anything is unnecessary, then clean up them”, used to remove unnecessary code.
- Codex. 5.6 Luna, OpenAI, prompt: “only use a SINGLE number and character in maps”, used to simplify map encoding.
- Codex. 5.6 Luna, OpenAI, prompt: “add a feature for blocks so that we can set it to be breakable or not...”, used to implement breakable blocks.
- Codex. 5.6 Luna, OpenAI, prompt: “when landed after falling ~25 blocks, the blocks weren’t broken”, used to fix high-impact block breaking.
- Codex. 5.6 Luna, OpenAI, prompt: “when breaking a block create the effect of that block shattering like the original mario”, used to add block-shattering effects.
- Codex. 5.6 Luna, OpenAI, prompt: “in addition to breakable blocks, add breakable pipes too.”, used to make pipes breakable.
- Codex. 5.6 Luna, OpenAI, prompt: “add mega mushroom to the game and mega state for the player...”, used to add the Mega Mushroom and Mega State.
- Codex. 5.6 Luna, OpenAI, prompt: “1. Make the mushroom bouncing around like the star... 2. When the player touch the mushroom, it starts transforming...”, used to refine Mega Mushroom interactions.
- Codex. 5.6 Luna, OpenAI, prompt: “1. no sparkle for mushroom... 4. when the pipe segment broken, delete the hitbox too...”, used to refine Mega State effects.
- Codex. 5.6 Luna, OpenAI, prompt: “Pipe segment hitboxes weren’t physically destroyed when broken, fix it”, used to remove destroyed pipe hitboxes.
- Codex. 5.6 Luna, OpenAI, prompt: “add 1-up mushroom, that adds +1 extra life to total count...”, used to add the 1-Up Mushroom.
- Codex. 5.6 Luna, OpenAI, prompt: “when using a warped pipe, make the player dive down into the pipe and get up out of the target pipe, while freezing the world”, used to animate pipe travel and freeze gameplay.
- Codex. 5.6 Luna, OpenAI, prompt: “add 2 horizontal warped pipes and a pair of non-reversed, reversed vertical warped pipes into map 2”, used to expand Map 2 pipe content.
- Codex. 5.6 Luna, OpenAI, prompt: “fix the horizontal pipes... The pipe should be entirely on top of the ground.”, used to correct horizontal-pipe placement.
- Codex. 5.6 Luna, OpenAI, prompt: “to enter a horizontal pipe, just move into it, no need for Interact”, used to simplify horizontal-pipe entry.
- Codex. 5.6 Luna, OpenAI, prompt: “when get in or out of the pipe, play the “pipe.mp3” sound”, used to add pipe sound effects.
- Codex. 5.6 Luna, OpenAI, prompt: “when player transform to another

form, play power_up.mp3…", used to add transformation and extra-life sounds.
- Codex. 5.6 Luna, OpenAI, prompt: "play starman_theme.mp3 in mega starman state and mega state", used to add special background music.
- Codex. 5.6 Luna, OpenAI, prompt: "fire state and super state when get hit will play transformation to normal state + play power_down.mp3", used to add power-down transformations.
- Codex. 5.6 Luna, OpenAI, prompt: "when shoot fireball, play fireball.mp3 Also, add particles for fireball too", used to add fireball audio and particles.
- Codex. 5.6 Luna, OpenAI, prompt: "when the player walking, play footstep.mp3 when the player jumping, play jump.mp3 when the player die, play dead.mp3 when the player run out of lives and defeated, play game_over.mp3", used to add player movement, death, and game-over sounds.
- Codex. 5.6 Luna, OpenAI, prompt: "when dying, wait for the dead sound to be almost finished then revive", used to synchronize death audio and respawning.
- Codex. 5.6 Luna, OpenAI, prompt: "why the sound only played when it finished itself before?… Fix it, also fix it with other sounds too", used to allow immediate and overlapping sound playback.
- Codex. 5.6 Luna, OpenAI, prompt: "when breaking blocks and pipes, play break.mp3 when killing an enemy, play kill.mp3", used to add breaking and enemy-kill sounds.
- Codex. 5.6 Luna, OpenAI, prompt: "when hit the lucky block, if there is power up, make it sprouting out of the lucky block and play sprout.mp3", used to animate power-ups emerging from blocks.
- Codex. 5.6 Luna, OpenAI, prompt: "Add feature to build maps inside the game, including add enemies, players, objects, blocks, … in each cells of the map, and save it into files so that we can play it later…", used to implement the in-game map editor.
- Codex. 5.6 Luna, OpenAI, prompt: "1. Make the map bigger, using mouse wheel to zoom in/out… Hold left mouse to add multiple blocks…", used to add zooming, panning, and bulk editing.
- Codex. 5.6 Luna, OpenAI, prompt: "1. zoom out is good but make the camera move outside the map edges too… 2. Ctrl + Z to undo, Ctrl + Y to redo…", used to improve editor navigation and controls.
- Codex. 5.6 Luna, OpenAI, prompt: "2. Instead of letters, place the actual sprite to the map… 3. For the lucky blocks, add parameters… 4. For the pipes, add all of its parameters…", used to improve object placement and configuration.
- Codex. 5.6 Luna, OpenAI, prompt: "change the name of the transparent_coin_block_spritesheet.png… All of the brick and coin block must have animation…", used to update block textures and animation.
- Codex. 5.6 Luna, OpenAI, prompt: "make the animation of bricks and coin blocks synced", used to synchronize block animations.
- Codex. 5.6 Luna, OpenAI, prompt: "increase maximum pipe warp id to

20”, used to expand warp-ID capacity.

- Codex. 5.6 Luna, OpenAI, prompt: “Add save/load game feature only for the default levels only - not the minigames. . . ”, used to implement save/load functionality.
- Codex. 5.6 Luna, OpenAI, prompt: “Bug: 1. Can not input the name of the slot 2. The used coin block texture after loading is not working. . . ”, used to fix save-slot naming and restored textures.
- Codex. 5.6 Luna, OpenAI, prompt: “when in save screen, use keyboard + arrow buttons and lock the mouse”, used to improve save-screen controls.

Sub-Room Warps, Map 1 Expansion & Gameplay Polish - Google Antigravity. Gemini 2.5 Pro, Google DeepMind, prompt: “please do not use any slopes”, used to remove slope tiles from Level 1 and ensure all ground surfaces use flat terrain. - Google Antigravity. Gemini 2.5 Pro, Google DeepMind, prompt: “please do not let the pipe in the air it just be placed stick into the ground and the grassland can not be in the air too”, used to ground all pipe obstacles directly onto the terrain floor and ensure all grassland tiles extend down to the bottom without floating in mid-air. - Google Antigravity. Gemini 2.5 Pro, Google DeepMind, prompt: “please fix that when the mario in the mega state, it can still break the unbreakable bricks”, used to configure Block collision in MegaState so that Mega Mario can crush environment blocks and pipes regardless of default breakability settings. - Google Antigravity. Gemini 2.5 Pro, Google DeepMind, prompt: “the game crashed”, used to diagnose and fix invalid spawn specifications in level JSON files. - Google Antigravity. Gemini 2.5 Pro, Google DeepMind, prompt: “can you please add the pipe that can move into a place that can earn many coins”, used to design an underground bonus coin vault connected via warp pipes. - Google Antigravity. Gemini 2.5 Pro, Google DeepMind, prompt: “please readd some piranhna plant in pipes into the map, and i mean add the pipe for teleporting is move into other environment not in the same map”, used to restore animated Piranha Plant pipe hazards and design an authentic sub-room warp architecture loading separate map files. - Google Antigravity. Gemini 2.5 Pro, Google DeepMind, prompt: “wait i want it to enter another environment not change the map from this level into other level”, used to implement executeSubRoomWarp in InGameScene and GameWorld, allowing cross-map pipe travel to map-1-bonus.json while preserving Level 1 HUD, score, timer, remaining lives, and power-up state. - Google Antigravity. Gemini 2.5 Pro, Google DeepMind, prompt: “why the map for level 1 has too many errors here, fix them”, used to resolve camera bounding conflicts, remove artificial background stretching, and restore standard 14-row level height. - Google Antigravity. Gemini 2.5 Pro, Google DeepMind, prompt: “why i entered the pipe, it appeared this”, used to diagnose missing player spawn tokens in map-1-bonus.json causing instant Game Over, add player spawns beside the arrival pipe, and add smooth pipe dive/rise delay timing. - Google Antigravity. Gemini 2.5 Pro, Google DeepMind, prompt: “why there are invisible bricks in the bonus map in map1, and it could not work normally”, used to diagnose invisible block collision caused by custom dynamic block prefabs, resolving it by using standard static brick tiles with automatic underground theme skinning. - Google Antigravity. Gemini 2.5 Pro, Google DeepMind, prompt: “cam you please make the map 1 longer”, used to expand Map 1 to 500 columns with new sections including Ravine Pit 6, aerial stepping stones, mid-air Mega Coins, double Piranha pipe gates, a third checkpoint, and a 10-step grand staircase. - Google Antigravity. Gemini 2.5 Pro, Google DeepMind, prompt: “please recheck and make sure that all your modifications do not effect and cause any errors in other parts in my whole project”, used to run comprehensive validation across all level files, prefabs, and minigame maps to prevent regressions.